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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Artificial Intelligence

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

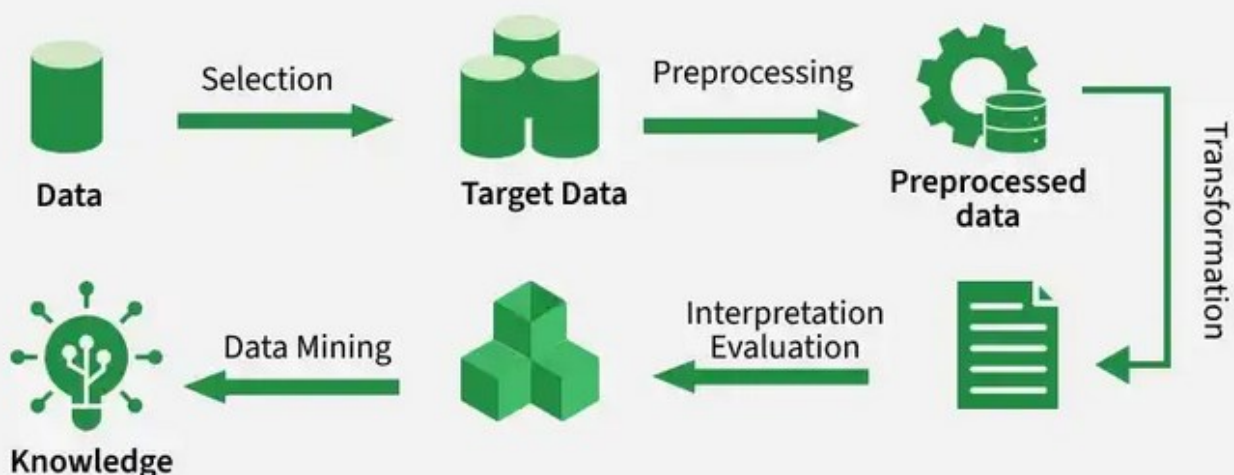
UNIT-02

Data Mining: Data Mining Functionalities, Data Pre-processing, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation, Architecture of A Typical Data Mining Systems, Classification of Data Mining Systems.

Data Mining:

- Data mining is the process of extracting insights from large datasets using statistical and computational techniques.
- It can involve structured, semi-structured or unstructured data stored in databases, data warehouses or data lakes.
- The goal is to uncover hidden patterns and relationships to support informed decision-making and predictions using methods like clustering, classification, regression and anomaly detection.
- Data mining is widely used in industries such as marketing, finance, healthcare and telecommunications.
- For example, it helps identify customer segments in marketing or detect disease risk factors in healthcare. However, it also raises ethical concerns particularly regarding privacy and the misuse of personal data, requiring careful safeguards.

Work flow of Data Mining



Difference Between Data Warehouse and Data Mining:

Feature	Data Warehouse	Data Mining
Definition	Stores and organizes large data for analysis.	Extracts patterns and insights from stored data.
Purpose	Helps businesses manage and retrieve structured data.	Helps analyze data for decision-making.
Process	Involves building a data warehouse, ETL, and reporting.	Uses data mining process like clustering and classification.
Tools	Data warehouse tools like Redshift, BigQuery, and Snowflake.	Uses AI, machine learning, and statistical methods.
Challenges	High setup cost, maintenance, and data integration issues.	Data quality issues, security concerns, and processing time.
Examples	Retail stores storing sales data for reports.	E-Commerce, predicting customer buying behavior.
Applications	Application of data warehouse: Business intelligence, reporting.	Data mining applications: Fraud detection, recommendations.
Outcome	Structured storage for better data management.	Smart insights for business growth.

Data mining functionalities:

Data mining functionalities are used to represent the type of patterns that have to be discovered in data mining tasks. In general, data mining tasks can be classified into two types including descriptive and predictive. Descriptive mining tasks define the common features of the data in the database and the predictive mining tasks act inference on the current information to develop predictions.

There are various data mining functionalities which are as follows –

- **Data characterization** – It is a summarization of the general characteristics of an object class of data. The data corresponding to the user-specified class is generally collected by a database query. The output of data characterization can be presented in multiple forms.
- **Data discrimination** – It is a comparison of the general characteristics of target class data objects with the general characteristics of objects from one or a set of contrasting classes. The

target and contrasting classes can be represented by the user, and the equivalent data objects fetched through database queries.

- **Association Analysis** – It analyses the set of items that generally occur together in a transactional dataset. There are two parameters that are used for determining the association rules –
 - It provides which identifies the common item set in the database.
 - Confidence is the conditional probability that an item occurs in a transaction when another item occurs.
- **Classification** – Classification is the procedure of discovering a model that represents and distinguishes data classes or concepts, for the objective of being able to use the model to predict the class of objects whose class label is anonymous. The derived model is established on the analysis of a set of training data (i.e., data objects whose class label is common).
- **Prediction** – It defines predict some unavailable data values or pending trends. An object can be anticipated based on the attribute values of the object and attribute values of the classes. It can be a prediction of missing numerical values or increase/decrease trends in time-related information.
- **Clustering** – It is similar to classification but the classes are not predefined. The classes are represented by data attributes. It is unsupervised learning. The objects are clustered or grouped, depends on the principle of maximizing the intra-class similarity and minimizing the intra-class similarity.
- **Outlier analysis** – Outliers are data elements that cannot be grouped in a given class or cluster. These are the data objects which have multiple behaviour from the general behaviour of other data objects. The analysis of this type of data can be essential to mine the knowledge.
- **Evolution analysis** – It defines the trends for objects whose behaviour changes over some time.

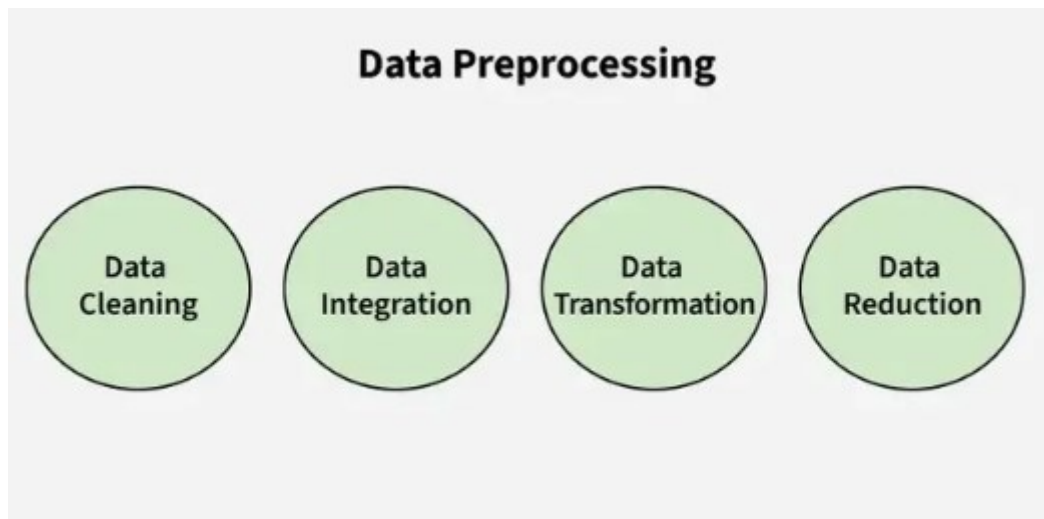
Data Pre-processing in Data Mining

Data pre-processing is the process of preparing raw data for analysis by cleaning and transforming it into a usable format. In data mining it refers to preparing raw data for mining by performing tasks like cleaning, transforming, and organizing it into a format suitable for mining algorithms.

- Goal is to improve the quality of the data.
- Helps in handling missing values, removing duplicates, and normalizing data.
- Ensures the accuracy and consistency of the dataset.

Steps in Data Pre-processing

Some key steps in data pre-processing are Data Cleaning, Data Integration, Data Transformation, and Data Reduction.



1. Data Cleaning: It is the process of identifying and correcting errors or inconsistencies in the dataset. It involves handling missing values, removing duplicates, and correcting incorrect or outlier data to ensure the dataset is accurate and reliable. Clean data is essential for effective analysis, as it improves the quality of results and enhances the performance of data models.

- **Missing Values:** This occurs when data is absent from a dataset. You can either ignore the rows with missing data or fill the gaps manually, with the attribute mean, or by using the most probable value. This ensures the dataset remains accurate and complete for analysis.
- **Noisy Data:** It refers to irrelevant or incorrect data that is difficult for machines to interpret, often caused by errors in data collection or entry. It can be handled in several ways:
 - **Binning Method:** The data is sorted into equal segments, and each segment is smoothed by replacing values with the mean or boundary values.
 - **Regression:** Data can be smoothed by fitting it to a regression function, either linear or multiple, to predict values.
 - **Clustering:** This method groups similar data points together, with outliers either being undetected or falling outside the clusters. These techniques help remove noise and improve data quality.
- **Removing Duplicates:** It involves identifying and eliminating repeated data entries to ensure accuracy and consistency in the dataset. This process prevents errors and ensures reliable analysis by keeping only unique records.

2. Data Integration: It involves merging data from various sources into a single, unified dataset. It can be challenging due to differences in data formats, structures, and meanings. Techniques like record linkage and data fusion help in combining data efficiently, ensuring consistency and accuracy.

- **Record Linkage** is the process of identifying and matching records from different datasets that refer to the same entity, even if they are represented differently. It helps in combining data from various sources by finding corresponding records based on common identifiers or attributes.
- **Data Fusion** involves combining data from multiple sources to create a more comprehensive and accurate dataset. It integrates information that may be inconsistent or incomplete from different sources, ensuring a unified and richer dataset for analysis.

3. Data Transformation: It involves converting data into a format suitable for analysis. Common techniques include normalization, which scales data to a common range; standardization, which adjusts data to have zero mean and unit variance; and discretization, which converts continuous data into discrete categories. These techniques help prepare the data for more accurate analysis.

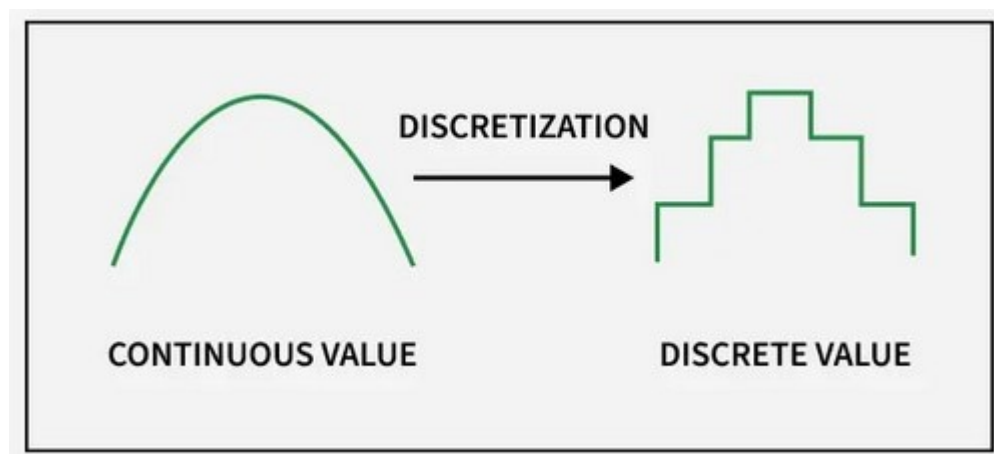
- **Data Normalization:** The process of scaling data to a common range to ensure consistency across variables.
- **Discretization:** Converting continuous data into discrete categories for easier analysis.
- **Data Aggregation:** Combining multiple data points into a summary form, such as averages or totals, to simplify analysis.
- **Concept Hierarchy Generation:** Organizing data into a hierarchy of concepts to provide a higher-level view for better understanding and analysis.

4. Data Reduction: It reduces the dataset's size while maintaining key information. This can be done through feature selection, which chooses the most relevant features, and feature extraction, which transforms the data into a lower-dimensional space while preserving important details. It uses various reduction techniques such as,

- **Dimensionality Reduction (e.g., Principal Component Analysis):** A technique that reduces the number of variables in a dataset while retaining its essential information.
- **Numerosity Reduction:** Reducing the number of data points by methods like sampling to simplify the dataset without losing critical patterns.
- **Data Compression:** Reducing the size of data by encoding it in a more compact form, making it easier to store and process.

Discretization

Discretization is the process of converting continuous data or numerical values into discrete categories or bins. This technique is often used in data analysis and machine learning to simplify complex data and make it easier to analyze and work with. Instead of dealing with exact values, discretization groups the data into ranges and helps algorithms perform better especially in classification tasks.

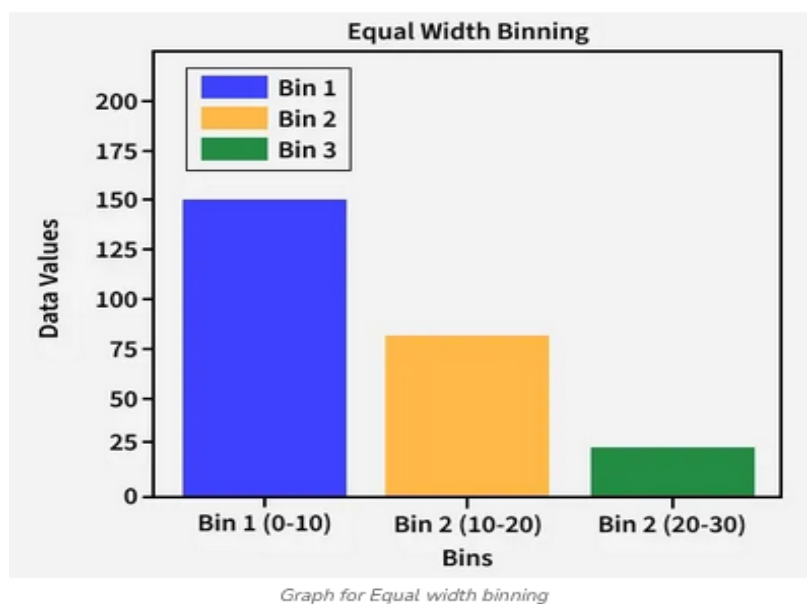


Types of Discretization Techniques

There are several types of discretization techniques used in data analysis to convert continuous data into discrete categories but mainly binning is used. Here are some of the common methods:

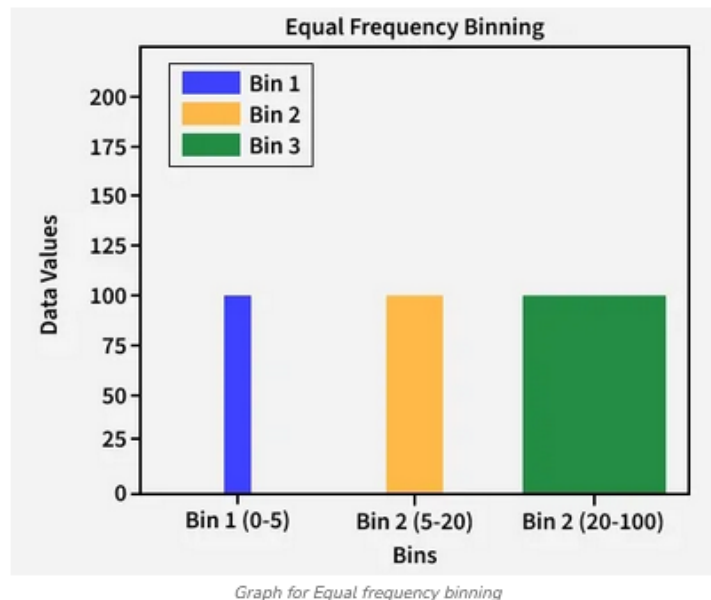
1. Equal Width Binning

- This technique divides the entire range of data into equal-sized intervals. Each bin has an equal width, determined by dividing the range of the data into n intervals.
- **Formula:** $\text{Bin Width} = \frac{\text{Max Value} - \text{Min}}{n}$
- For example, if you have data from 1 to 100, you can divide it into 5 intervals: 1-20, 21-40, 41-60, 61-80, and 81-100.
- It is best for Simple datasets with uniform distribution.



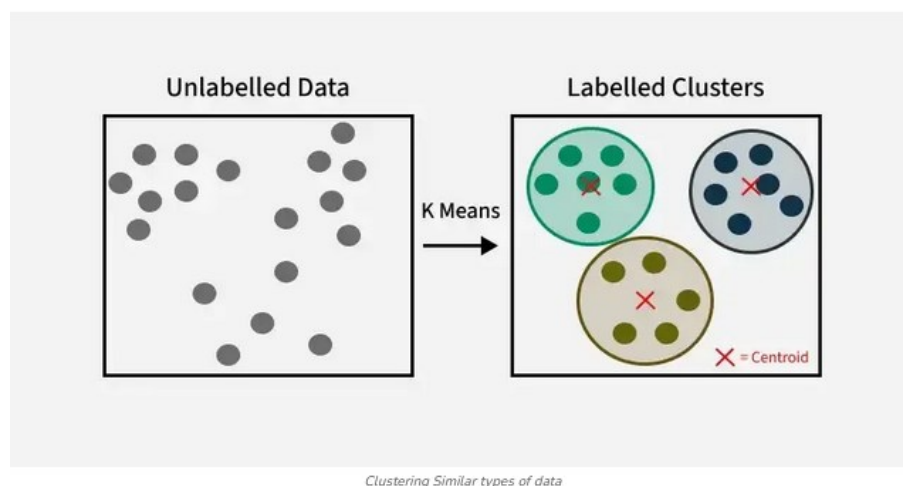
2. Equal Frequency Binning

- Data is divided so that each bin contains approximately the same number of values.
- This method divides the data so that each interval has the same number of data points.
- For example, if you have 100 data points, you might divide them into 5 intervals, each containing 20 data points.
- It is best for Skewed or non-uniform distributions.



3. K-means Clustering

- To create intervals or bins that group similar observations, we can use clustering algorithms like k-means.
- In discretization using k-means clustering, the partitions are the clusters identified by the k-means algorithm.
- K-means Clustering uses clustering algorithms to group data into clusters based on similarity. The data points in each cluster are treated as a single category.
- It is best for Complex datasets where natural groupings exist.



4. Decision Tree Discretization

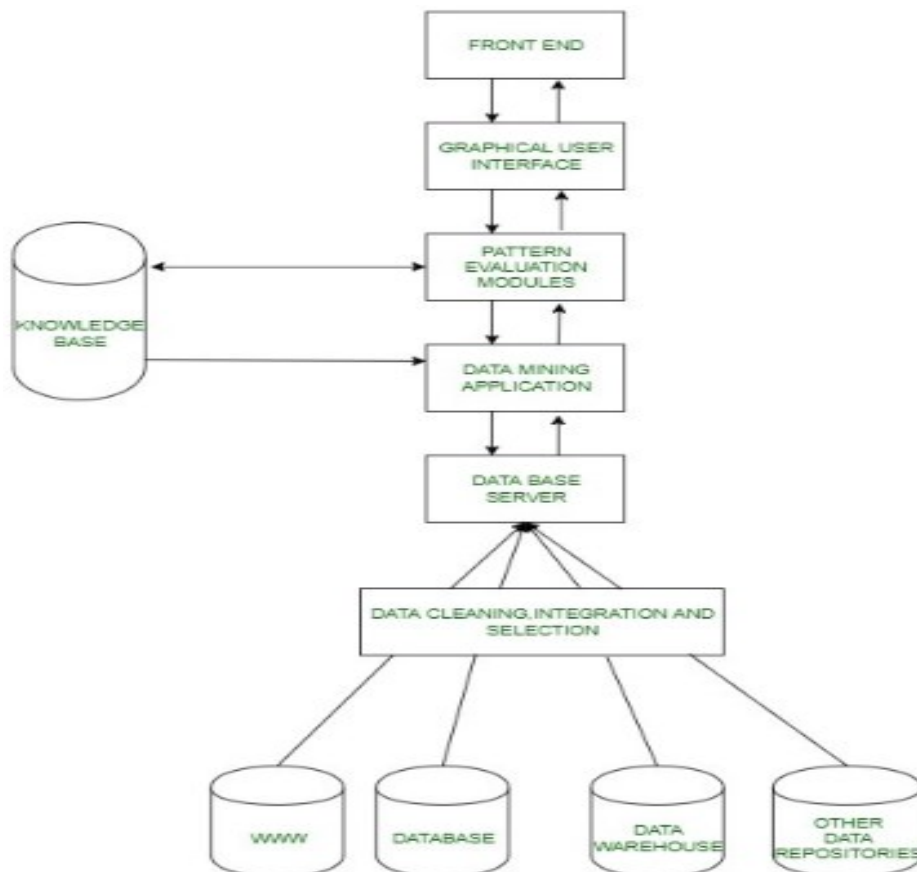
- Decision tree methods discretize continuous attributes during the learning process.
- This method uses decision trees to split the data based on feature values, turning continuous variables into discrete categories that help in prediction.
- There are several decision tree algorithms that are designed for specific tasks such as classification, regression, or handling imbalanced data.
- Discretization with decision trees is another top-down approach that consists of using a decision tree to identify the optimal partitions for each continuous variable.



5. Custom Binning

- Domains such as healthcare, finance, or demographics may require manually defined bins.
- This method relies on domain knowledge and allows for highly interpretable categories.
- **Example:** Categorizing age into ranges like 0–18, 19–40, and 41+.

The architecture of Data Mining



Basic Working:

1. It all starts when the user puts up certain data mining requests, these requests are then sent to data mining engines for pattern evaluation.
2. These applications try to find the solution to the query using the already present database.
3. The metadata then extracted is sent for proper analysis to the data mining engine which sometimes interacts with pattern evaluation modules to determine the result.
4. This result is then sent to the front end in an easily understandable manner using a suitable interface.

A detailed description of parts of data mining architecture is shown:

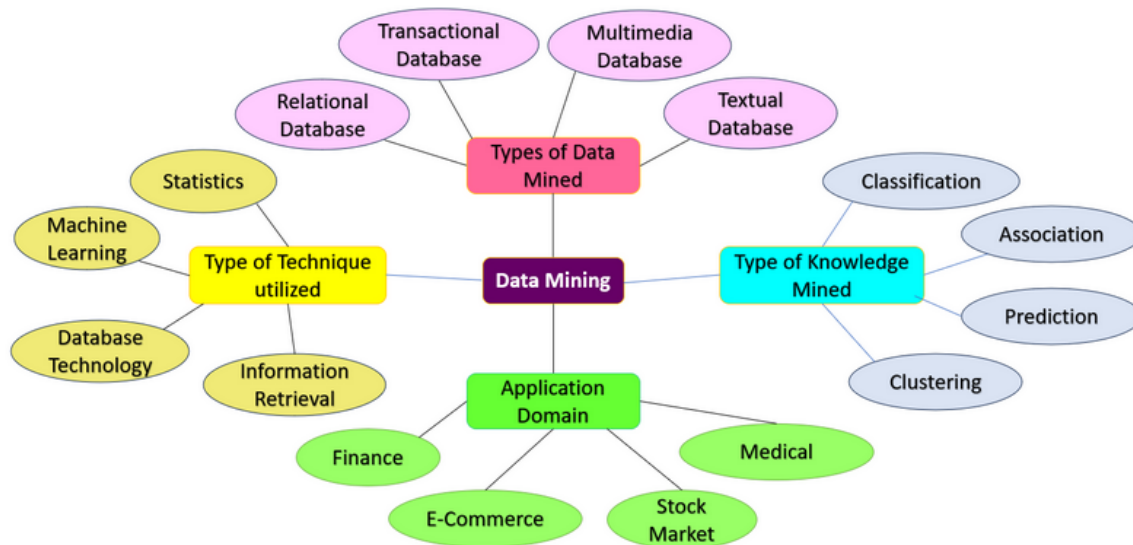
1. **Data Sources:** Database, World Wide Web(WWW), and data warehouse are parts of data sources. The data in these sources may be in the form of plain text, spreadsheets, or other forms of media like photos or videos. WWW is one of the biggest sources of data.
2. **Database Server:** The database server contains the actual data ready to be processed. It performs the task of handling data retrieval as per the request of the user.
3. **Data Mining Engine:** It is one of the core components of the data mining architecture that performs all kinds of data mining techniques like association, classification, characterization, clustering, prediction, etc.
4. **Pattern Evaluation Modules:** They are responsible for finding interesting patterns in the data and sometimes they also interact with the database servers for producing the result of the user requests.
5. **Graphic User Interface:** Since the user cannot fully understand the complexity of the data mining process so graphical user interface helps the user to communicate effectively with the data mining system.
6. **Knowledge Base:** Knowledge Base is an important part of the data mining engine that is quite beneficial in guiding the search for the result patterns. Data mining engines may also sometimes get inputs from the knowledge base. This knowledge base may contain data from user experiences. The objective of the knowledge base is to make the result more accurate and reliable.

Classification of Data Mining systems:

- Data Mining is considered as an interdisciplinary field. It includes a set of various disciplines such as statistics, database systems, machine learning, visualization, and information sciences.
- Classification of the data mining system helps users to understand the system and match their requirements with such systems.
- Data mining discovers patterns and extracts useful information from large datasets.
- Organizations need to analyze and interpret data using data mining systems as data grows rapidly. With an exponential increase in data, active data analysis is necessary to make sense of it all.

Data mining (DM) systems can be classified based on various factors.

- Classification based on Types of Data Mined
- Classification based on Type of knowledge Mined
- Classification based on Type of Technique Utilized
- Classification based on Application Domain



1) Classification based on Types of Data Mined:

A database mining system can be classified based on 'type of data' or 'use of data' model or 'application of data.'

This classification is determined by the format and source of the data. While most people think of spreadsheets, data mining can handle much more complex structures.

For Example: Relational Database, Transactional Database, Multimedia Database, Textual Data, World Wide Web (WWW) and etc.

- **Relational Databases:** Mining structured data stored in tables.
- **Transactional Data:** Analyzing logs of sales or "market baskets" (e.g., what items are frequently bought together).

2) Classification based on Type of knowledge Mined:

This looks at the **goal** of the mining process. What kind of "answers" are we looking for?

We can classify a data mining system according to the kind of knowledge mined. It means the data mining system is classified based on functionalities such as

- **Characterization & Discrimination:** Summarizing the features of a specific class (e.g., "What makes a loyal customer?") or comparing it against others.
- **Association Analysis:** Finding rules like "If a customer buys a laptop, they are 80% likely to buy a mouse."
- **Classification & Prediction:** Building models to categorize new data (e.g., flagging an email as "Spam") or predicting future values.
- **Clustering:** Grouping data points that are similar to each other without having pre-defined labels.
- **Outlier Analysis:** Identifying "anomalies" or data points that don't fit the pattern (essential for fraud detection).

3) Classification based on Type of Technique Utilized:

We can classify a data mining system according to the kind of techniques used. We can describe these techniques according to the degree of user interaction involved or the methods of analysis employed.

Data Mining systems use various techniques, including Statistics, Machine Learning, Database Systems, Information retrieval, Visualization, and pattern recognition.

This category defines the **methodology** or the "brain" behind the mining.

- **Machine Learning:** Using algorithms like Neural Networks or Decision Trees that "learn" from data.
- **Pattern Recognition:** Identifying shapes or recurring structures within the data.
- **Statistics:** Using mathematical models to measure data variance and significance.
- **Database-oriented Techniques:** Using high-level query languages or optimized indexing to find patterns within massive datasets.

4) Classification based on Application Domain:

Finally, systems are often tailored to the specific **industry** they serve, as the "rules" of the data change depending on the field.

We can classify a data mining system according to the applications adapted. These applications are as follows

- **Telecommunications:** Analyzing call patterns and network loads.
- **Retail/E-commerce:** Recommendation engines and inventory management.
- **Finance/Banking:** Credit scoring, risk assessment, and fraud detection.
- **Bioinformatics:** Analyzing DNA sequences and protein structures.

Concept Hierarchy in Data Mining:

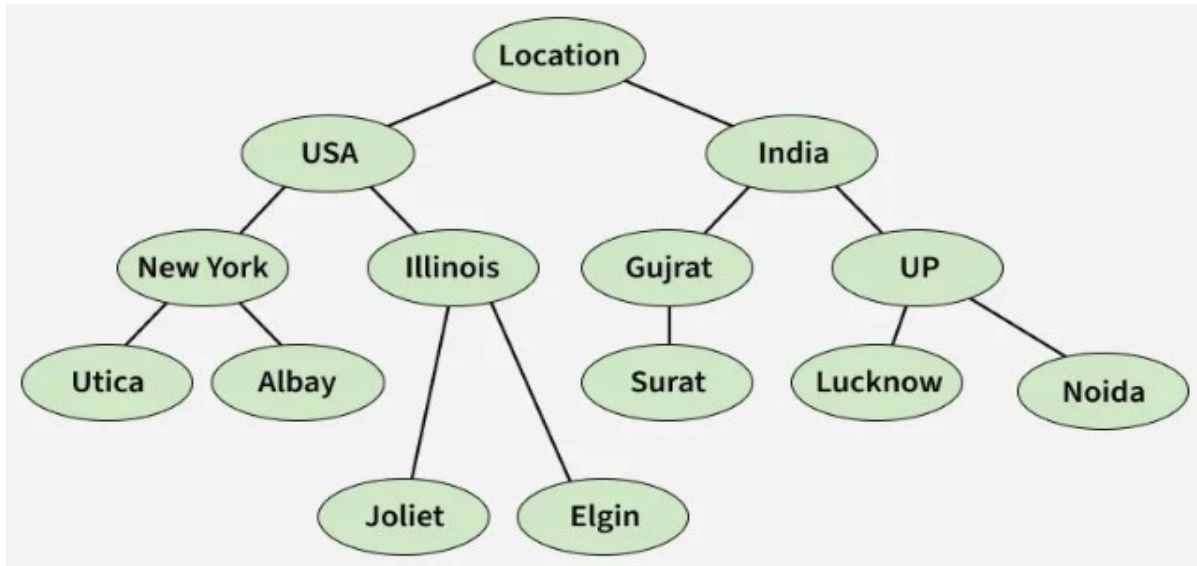
- In data mining, a concept hierarchy is a way of organizing data into different levels of abstraction.
- A concept hierarchy organizes data into multiple levels of abstraction-ranging from detailed (low-level) values to more general (high-level) concepts.
- It allows users to drill down for detailed analysis or roll up for summaries, helping simplify large datasets and improving pattern discovery.

Example of a Concept Hierarchy

The diagram below shows a hierarchy for the Location dimension.

- The root node Location is generalized into Countries (USA, India).
- Countries break into States (New York, Gujarat).
- States further break down into Cities.

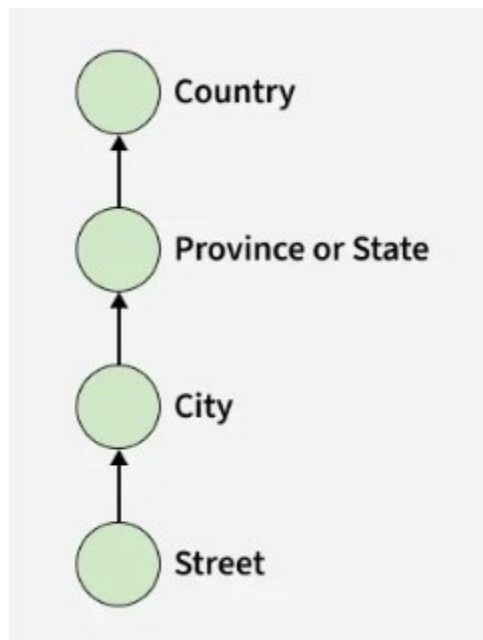
Such hierarchies allow a user to analyze data at the level they need—city-level detail or country-level summaries.



Concept Hierarchy Example

Concept Hierarchy Example

The hierarchical structure represents the abstraction level of the dimension location, which consists of various footprints of the dimension such as:



Types of Concept Hierarchies

Concept hierarchies can be created in different ways depending on the data and the domain:

1. Schema-Based Hierarchy:

- This hierarchy is derived directly from the database schema by utilizing existing relationships between attributes.

- It represents a total or partial order between attributes (e.g., primary key to foreign key).
- It Commonly used in data warehousing to integrate data from multiple sources into a unified structure.
- Derived from the database schema (e.g., primary key → foreign key relationships). Useful in data warehouses for structuring dimensions such as:
 - City → State → Country
 - Day → Month → Year

2. Set-Grouping Hierarchy:

- This type groups specific values or instances into categories based on set membership.
- Each set is defined by its members, allowing a "bottom-up" construction from low-level data points to higher-level concepts.
- **Example:** Grouping specific items like "Banana" and "Apple" into the "Fruit" set.
 - Example: Product categories → Sub-categories → Items
- It Highly effective for data cleaning, preprocessing, and managing outliers.

3. Operation-Derived Hierarchy:

- These hierarchies are generated by applying mathematical or statistical operations to the raw data.
- Operations like aggregation, normalization, or range partitioning are applied to transform data into more abstract views.
- Example: Age → Age Range (0–12 Child, 13–19 Teen, 20–60 Adult, >60 Senior)

Mapping an exact age (e.g., 25) to a category like "Adult" based on range partitioning (e.g., 20–60).

- It Frequently used in clustering and dimensionality reduction tasks.

4. Rule-Based Hierarchy:

- In this model, the hierarchy is defined through user-specified rules or conditional statements.
- Data points are assigned to classes dynamically based on whether they meet certain criteria.
- It is Essential for complex classification, decision-making, and expert systems.

- **Example:** A rule such as `IF annual_income > 1,000,000 THEN Category = "High Income"`.

Example rule:

- `IF income > 10,00,000 → High income; ELSE → Low/Medium income`

5. Better Representation of Domain Knowledge

- Concept hierarchies serve as a bridge between raw data and human understanding by incorporating specialized expertise.

- Unlike automated hierarchies, these capture real-world relationships that may not be obvious in the database schema alone.
- They allow users to view data from different perspectives and discover "hidden" patterns at the most relevant level of abstraction for their industry.
- **Example:** A doctor defining a hierarchy of symptoms to better identify a specific disease.

Hierarchies capture real-world relationships (e.g., Product → Brand → Category), making the data model easier to understand.